SRI SAI HEMANTH BOLLEPALLI

AI Researcher | Medical Imaging & Deep Learning

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SUMMARY

Master of Artificial Intelligence student at UTS focused on medical image segmentation, MRI analysis and computer vision. Participant and manuscript reviewer in the MICCAI 2026 BraTS-METS Challenge, with hands-on work across nnU-Net, transformers and LLM-based labelling. Seeking research roles building reliable, clinically useful deep learning systems.

RESEARCH & PROJECT EXPERIENCE

Brain Metastasis & Resection Cavity Segmentation

MICCAI 2026 BraTS-METS Challenge, continued as UTS research project | 2026 - Present

- Built an end-to-end segmentation pipeline for the MICCAI 2026 BraTS-METS Challenge (team Core_Neuro_AI), working independently.
- Trained customised nnU-Net v2 models on multi-modal 3D MRI (T1, T1c, T2, FLAIR) to segment 4 regions: WT, TC, ET and resection cavity.
- Extended the work as a UTS research project, benchmarking a CNN (nnU-Net ResEnc-L) against a transformer (Primus) and ensembling both with per-region threshold tuning.
- Reviewed 3 manuscripts for the challenge, assessing method, evaluation and reporting quality.

Knee MRI Abnormality Detection (RSNA)

University of Technology Sydney | 2026

- Co-developed a 2-person pipeline to detect 12 abnormality types from knee MRI scans.
- Built a radiology report labeller with a local Qwen LLM, reaching 0.858 macro AUC on 58 gold-standard reports and matching a regex + MedGemma fusion baseline.
- Kept compute cost near zero by moving training from GCP to Kaggle notebooks with a cached, preprocessed scan store.

Rare-Class-Aware Multimodal Skin Lesion Diagnosis

ISIC / MILK10k

- Built a PyTorch pipeline fusing paired clinical images, dermoscopic images and patient metadata.
- Improved rare-class recall with targeted oversampling and a ConvNeXt + EfficientNet ensemble.
- Packaged inference in Docker for reproducible, one-command deployment.

AeroTail: Deep RL Drone Target Following

University of Technology Sydney | 2026

- Trained an autonomous drone to visually track and follow a moving target in the AirSim simulator.
- Ran and compared SAC and PPO agents, then handed over results to drive a final RecurrentPPO experiment.

TRUST-ADMET: Trustworthy AI for ADMET Prediction

Open source

- Created a Python framework combining classical ML, graph neural networks and molecular transformers for drug property prediction.
- Added reliability diagnostics so users can see when a prediction should not be trusted.

SKILLS

Languages: Python, SQL
ML & Medical Imaging: Deep Learning, Computer Vision, Semantic Segmentation, Medical Image Analysis, Multi-modal MRI, Brain Tumour Segmentation, Reinforcement Learning, LLM-based Labelling
Frameworks: PyTorch, MONAI, nnU-Net v2, SimpleITK, NiBabel, Docker
MLOps & Platforms: Weights & Biases, ClearML, Linux, CUDA, HPC, Google Cloud Platform, Amazon SageMaker, Kaggle, RunPod
Research: Experimental Design, Model Evaluation, Literature Review, Scientific Writing, Peer Review, Reproducible Research

EDUCATION

Master of Artificial Intelligence - University of Technology Sydney | 2025 - 2027
Bachelor of Technology in Computer Science - Jawaharlal Nehru Technological University Kakinada | 2020 - 2024

CERTIFICATIONS

Microsoft Certified: Azure Fundamentals (AZ-900) | 2023
Google Cloud Architect Specialization, Coursera | 2022